

Department of Economics
Jahangirnagar University
4th Year (Hons) 2019
Course No-404 (Health Economics)
1st Tutorial Exam (Full Marks : 20)

Answer any **Two** questions: .

- ① (a) What is health economics? Does Economics apply to health and health care?
(b) Why health care is different from other goods and services? Explain.
- ② Using supply and demand graph and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following:
(a) An increase of the price of Ace on the market for Napa (Hint : Ace and Napa are substitutes).
(b) A price ceiling placed on physician fees on the market for physician services.
(c) Increased cost of medical education on the market for physician services.
(d) The virtual elimination of smoking in the population on the market for hospital services.
3. (a) Define health production function. Would the health production function eventually bend downward? Explain in support of your answer.
(b) Explain the two different theories of schooling.
4. (a) The role of medicine in the historical decline of the population mortality rate is small, perhaps negligible. Explain this statement with empirical evidence.
(b) According to Victor Fuch, what factors are responsible for the historical decline in the infant mortality rate? Explain. Does this mean medical research is unimportant? Explain.

Department of Economics
Jahangirnagar University
4th Year (Hons) Examination 2018
Course title: Health Economics
Course No. 404
1st tutorial Examination
Full Marks : 20



Answer any Two (02) questions of the following:

1. (a) Does economics apply to health and health care? Explain. 4
(b) Why medical care is different from other goods and services? Explain. 6
2. (a) If not medicine, then what factors were responsible for the historical decline in the population mortality rate? Discuss. 6
(b) Explain the two different theories of schooling. 4
3. (a) Define health production function. Would the health production function eventually bend downward? Explain. 6
(b) According to Victor Fuchs, what factors were responsible for the historical decline in the infant mortality rate in New York? Explain. 4
4. (a) According to Grossman model, how health care demand is different from traditional approach of demand? 4
(b) Explain home good function and health investment function in the light of Grossman model. 3
(c) How optimum health stock is determined? Explain with diagram. 3
5. (a) Explain, in the light of Grossman model, how an individual chooses optimum amount of income and leisure. 5
(b) How is it possible to earn more money and more leisure time together? Explain, in the light of Grossman model.

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Jahangirnagar University
Faculty of Social Sciences
Department of Economics
Health Economics
2nd and 3rd tutorial test

Full Marks : 40

All question carries equal marks: Answer any three (3) questions

1. (a). Suppose that priya's income is \$40,000 per year. She can spend it on health care visits, which cost \$80 per visit, or on groceries (standing for all other goods), which cost \$100 per bag of groceries. Draw priya's budget constraint. Using indifference curves, show Priya's optimum if she buys 200 bags of groceries per year.

(b) Suppose that priya's income rises to \$48,000 per year and that she increases her consumption of health care visits by 50 visits. Using the graphs for question 1 (a), draw the new equilibrium. what is her income elasticity of demand for health care visits? Interpret your result.

2. (a) How would you derive the demand curve for health care visits from the theory of consumer behavior?

(b) Explain the role of time and coinsurance on the demand for health care visits.

3. Discuss the determinants of market health care visit functions.

4. Suppose that the market demand for medical care is summarized by the demand function:

$$Q^d = 100 - 2p$$

and the market supply is summarized by the supply function:

$$Q^s = 20 + 2p$$

(a) Calculate the equilibrium price and quantity, assuming no health insurance is available.

(b) Suppose that health insurance is made available that provides for a 20 percent coinsurance rate. Calculate the new equilibrium price and quantity.

(c) Calculate the deadweight loss due to this insurance.

5. (a) According to Mark Pauly (1968) model, how optimum amount of insurance coverage is determined in the market?

(b) Using the same model, show the impact of the changes in premiums, expected loss, and wealth on the optimum amount of coverage.

6. (a) What is insurance? Define the following terminology of insurance:

Premium and coverage, coinsurance and copayment, deductible and limitations.

(b) "Insurance can be sold only in circumstances where there is diminishing marginal utility of wealth" Explain the above statement with the help of a diagram.

7. (a) Define asymmetric information what problems arise due to asymmetric information? Give examples.

(b) Using Akerlof (1970) model show that market will not exist if there is asymmetric information.

(c) Apply Akerlof model to explain health insurance market.

8. (a) What is supplier induced demand (SID)? Explain it with the help of a diagram.

(b) Explain the target income hypothesis of SID.

Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination 2018
Course No: 404 (Health Economics)
Time: 4 Hours Full Marks: 70

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Answer any FIVE of the following questions. All parts of a particular question must be answered consecutively

1. a) Define health economics. Why does the share of medical care to GDP increase in recent years? [6]
b) Why is medical care different from other goods and services? Explain. [8]

2. a) If not medicine, what factors were responsible for the historical decline in the population mortality rate? Explain. [8]
b) Explain the two different theories of schooling. [6]

3. a) Explain, in the light of Grossman model, how optimal health stock is determined? [4]
b) How does an individual investment in health change in response to change in age and education? [7]
c) What are the limitations of Grossman model? [3]

4. Consider demand curves for Aspirin, estimated for two different sets of consumers: [14]
(a) $Q = 20 - 5P + 0.2Y$
(b) $Q = 30 - 5P + 0.2Y$
If $Y = \$20$ and $P = \$1$, calculate the price and income elasticities for group (a) and group (b). Whose elasticities will be higher? Why?

5. a) Discuss the determinants of market health care visit functions. [10]
b) What is the role of time and coinsurance on the demand for health care? [4]

6. a) Explain the major problems that arise from asymmetric information. Give specific example. [4]
b) Using Akérlof model (1970), show that market will not exist if there is asymmetric information. [6]
c) Illustrate the application of Lemon principle to explain the health insurance market. [4]

7. a) According to Mark Pauly (1968) model, how optimum amount of insurance coverage is determined in the market? [4]
b) Using the same model, show the impact of the changes in premiums, expected loss and wealth on the optimum amount of coverage. [10]

8. a) Define Supplier Induced Demand (SID)? Explain it with the help of diagram. [8]
b) Explain the target income hypothesis of SID. [6]

Chapter - 1, 2, 5, 7

Department of Economics
Jahangirnagar University
4th Year (Hons) 2017
Course No-404 (Health Economics)
1st Tutorial Exam (Full Marks : 20)

Answer any Two questions:

1. (a) What is health economics? Explain the relevance of health economics. — P: 3
P-2

(b) Why health care is different from other goods and services? Explain. P-13

2. Using supply and demand graph and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following:

(a) A decrease of the price of Ace on the market for Napa (Hint : Ace and Napa are substitutes).

(b) Increased Price of painkillers on the market for antiulcerative drugs (Hint : Painkillers and antiulcerative drugs are complements).

(c) Increased cost of medical education on the market for physician services.

(d) The virtual elimination of smoking in the population on the market for hospital services.

3. (a) Define health production function. Would the health production function eventually bend downward? Explain in support of your answer. 85

(b) Explain the two different theories of schooling.

4. (a) According to Grossman model, how health care demand is different from traditional approach of demand? — 129

(b) Explain, in the light of Grossman model, the impact of age and education on the optimum health stock. 137

5. (a) How would you explain the potential use of an individual's time by labor-leisure trade off? show the equilibrium of an individual's time by labor-leisure line and indifference curve. 131 - 133

(b) Is it possible to earn more income and to take more leisure time together? Explain in the light of Grossman model.

6. (a) The role of medicine in the historical decline of the population mortality rate is small, perhaps negligible. Explain this statement with empirical evidence. 88

(b) According to Victor Fuch, what factors are responsible for the historical decline in the infant mortality rate? Explain. Does this mean medical research is unimportant? Explain. 91

Answer one (1) from Part A and another one (1) from Part B.

Part A (Consumer Choice and Demand) - 9

1 (a). Suppose that priya's income is \$40,000 per year. She can spend it on health care visits, which cost \$80 per visit, or on groceries (standing for all other goods), which cost \$100 per bag of groceries. Draw priya's budget constraint. Using indifference curves, show Priya's optimum if she buys 200 bags of groceries per year.

(b) Suppose that priya's income rises to \$48,000 per year and that she increases her consumption of health care visits by 50 visits. Using the graphs for question 1 (a), draw the new equilibrium. what is her income elasticity of demand for health care visits? Interpret your result.

2. (a) How would you derive the demand curve for health care visits from the theory of consumer behavior? 175

(b) Explain the role of time and coinsurance on the demand for health care visits. 178

3. Discuss the determinants of market health care visit functions. 187

Part B (Health Insurance) 8

4. Suppose that the market demand for medical care is summarized by the demand function:

$$Q^d = 100 - 2p$$

and the market supply is summarized by the supply function:

$$Q^s = 20 + 2p$$

(a) Calculate the equilibrium price and quantity, assuming no health insurance is available.

(b) Suppose that health insurance is made available that provides for a 20 percent coinsurance rate. Calculate the new equilibrium price and quantity.

(c) Calculate the deadweight loss due to this insurance.

5. (a) According to Mark Pauly (1968) model, how optimum amount of insurance coverage is determined in the market? 163

(b) Using the same model, show the impact of the changes in premiums, expected loss, and wealth on the optimum amount of coverage. 155

6.(a) What is insurance? Define the following terminology of insurance: 170

Premium and coverage, coinsurance and copayment, deductible and limitations. 140-150

(b) "Insurance can be sold only in circumstances where there is diminishing marginal utility of wealth" Explain the above statement with the help of a diagram.

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4th Year (Hons) 2017
3rd Tutorial Examination
Course No. Econ (404) Health Economics

Answer question No. 1 (Compulsory) and another one question from questions 2 to 4.

1. What is asymmetric information? Show, with the help of used car market example introduced by George Akerlof (1970), that market will not exist if there is asymmetric information. Show also that market will exist if there is symmetric information. Apply Akerlof model to explain the health insurance market. 196-199
2. a) Define cost minimization analysis (CMA), cost effectiveness analysis (CEA), cost utility analysis (CUA) and cost benefit analysis (CBA).
b) Consider a health care project that costs 10,000 today. It will provide benefits of \$4000 at the end of Year 1, \$ 3500 at the end of year 2 and \$ 3500 at the end of year 3. If the discount rate is 6%, will this project be approved using CBA? Would Your answer change if the discount rate is 5%? 4%?
3. a) What is discounting? why do we Discount future value into its present value? Explain.
(b) How do we value human life? Explain. 72
4. Explain the basic principles of Cost benefit analysis (CBA).-66

Answer any FIVE of the following questions. All parts of a question must be answered consecutively.

1. a) Why do we study health economics? Does economics apply to health and health care? [7]
 b) Is health care different? Explain. [7]

2. a) Define production function. Describe the production function of health. [6]
 b) Would health production function eventually bend downward? Explain your view point. [8]

3. a) Explain health investment function and home good function in the light of Grossman model. [6]
 b) Explain the impact of age, wage and education on optimal health stock. T_H [8]

4. a) How would you derive the demand for health care visit from the theory of consumer choice? T_B [8]
 b) Explain the role of time and coinsurance on the demand for health care visit. [6]

5. Suppose that the market demand for medical care is summarized by the demand function:

$$Q_d = 100 - 2P$$

and the market supply is summarized by the supply function:

$$Q_s = 20 + 2P$$

- a) Calculate the equilibrium price and quantity, assuming no health insurance is available. [4]
 b) Suppose that health insurance is made available that provides for a 20% coinsurance rate. Calculate the new equilibrium price and quantity. [5]
 c) Calculate the deadweight loss due to this insurance. [5]

6. a) What is asymmetric information? Using Akerlof model (1970), show that market will not exist if there is asymmetric information. [8]

- b) Apply the Akerlof model to explain the health insurance market. [6]

7. a) Define cost benefit analysis (CBA). Explain the decision criteria of CBA. [5]
 b) Why do we discount future value into their present value? Explain. [4]
 c) How do we value life? Explain. [3]

8. a) What is Supplier Induced Demand (SID)? Explain it with the help of diagram. [8]

b) Explain the target income hypothesis of SID.

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Potential,
Feeling
Future. - 2008 - 2010

Department of Economics
Jahangirnagar University
4th Year (Hons.) Final Examination 2016
Course No: 404 (Health Economics)
Time: 4 Hours Full Marks: 70

Answer any five of the following questions. All parts of a question must be answered consecutively

1. a) Define health economics. Does economics apply to health and health care? [6]
b) Why is medical care different from other goods and services? Explain. [8]
2. a) Define cost minimization analysis (CMA), cost effectiveness analysis (CEA), cost utility analysis (CUA), and cost benefit analysis (CBA). [6]
b) Explain the basic principles of CBA. [8]
3. a) "The historical role of medicine in the decline of the population mortality rate is very small, perhaps negligible"-Explain the above statement with practical examples. [8]
b) Explain the two different theories of schooling. [6]
4. a) Explain the potential uses of an individual's time by labor-leisure trade-off? [6]
b) Is it possible to earn more money and more leisure time together? Explain the light of Grossman model. [8]
5. a) How would you derive the demand curve of health care visit from the theory of consumer choice? [8]
b) Explain the role of time and coinsurance on the demand for health care visits. [6]
6. a) Define the following terminology of insurance: [6]
i) Premium and coverage
ii) Coinsurance and copayment, and
iii) Deductibles
b) Explain the effect of an increase in premium rate and expected loss on the optimum health insurance coverage. [8]
7. a) Explain the Akerlof (1970) model of asymmetric information through an analysis of the used car market. [8]
b) Explain health insurance market by the help of Akerlof model. [6]
8. a) Define the term "Economic Evaluation". Briefly explain why economic evaluation is important for the health sector. — 40 [4]
b) Identify various components of a full economic evaluation. [4]
c) Critically discuss the "Cost Minimization Analysis (CMA)" with one illustrative and one practical example. [6]

Health Economics

Answers to any two questions of the following using a supply and demand graph.

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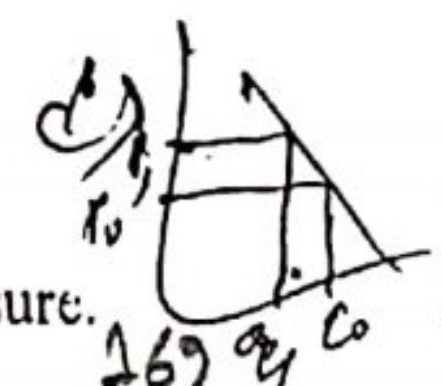
404

Answer any two questions of the following:

Using a supply and demand graph and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following:

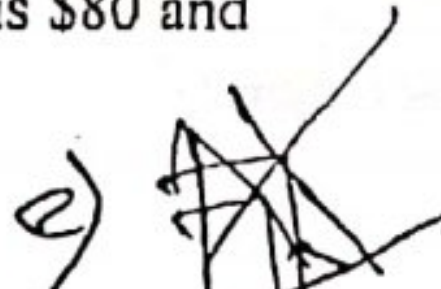
- (a) Increased price of Azith on the market for Zimax (Azith and Zimax are substitutes).
- (b) Increased graduations of new doctors on the market for physician services.
- (c) Decreased price of pain killers on the market for anti ulcerative drugs. (complement)
- (d) Increased cost of medical education on the market for physician services.
- (e) The virtual elimination of smoking in the population on the market for hospital services.

- (a) Explain the health investment function and home good function in the light of Grossman model. — 167 (chapter 17)
- (b) How would you explain the potential use of a person's time by the labor-leisure trade-off? 167
- (c) Explain the impact of an increase in activities to improve health on income and leisure.



Suppose Andrew's income is \$40,000 and he can spend this amount on visits or groceries (stands for all other goods) or a combination of both. The cost of a visit is \$80 and the price of a bag of grocery is \$100.

- (a) Make a table showing at least six consumption bundles (including two extreme consumption bundles) that Andrew can afford.
- (b) Draw the budget line.
- (c) Suppose Q_v is the quantity of visit and Q_g is the quantity of grocery. Write down the budget equation.
- (d) Define opportunity cost. What is the opportunity cost of having five additional visits?
- (e) Suppose Andrew's utility function is $U = Q_v Q_g$. Using the Lagrangian method, find the utility maximizing number of visits and groceries. And also find the maximum utility of utility.
- (f) Find the marginal utility of money. Interpret your result.



- (a) Define production function. Describe the production function of health. Would the production function of health eventually bend downward? Give reasons in support of your answer. 117
- (b) "Practitioner-provided medical interventions played a small, perhaps negligible, role in the historical decline in population mortality rates." Explain this statement with the help of empirical evidence. 122

- (a) Define price and income elasticity of demand for health care. → (cases for two sets of consumers)
- (b) Consider demand curves for aspirin, estimated for two different sets of consumers:
 - (i) $Q = 20 - 5P + 0.2Y$
 - (ii) $Q = 30 - 5P + 0.2Y$

If $Y = \$20$ and $P = \$1$, Calculate the price and income elasticities for group (i) and group (ii). Whose elasticities will be higher? Why?

$Q_1 = 19$

→ बाकि आह

(ii) over any two out of the three

Sharon Blawie
Dept. of Economics

Answer any two of the following questions

Answer any two of the following questions

1. Suppose that the market demand for medical care is summarized by the demand function:
 $Q_d = 100 - 2p$
and the market supply is summarized by the supply function:
 $Q_s = 20 + 2p$

- Graph the demand and supply function in one diagram.
- Calculate the equilibrium price and quantity, assuming no health insurance is available.
- Suppose that health insurance is made available that provides for a 50 percent coinsurance rate. Calculate the new equilibrium price and quantity.
- Calculate and indicate the deadweight loss in the diagram due to this insurance.

2. (a) Define cost benefit analysis(CBA). Explain the basic principles of CBA.
(b) Consider a health care project that costs \$10,000 today. It will provide benefits of \$4,000 at the end of year 1, \$3,500 at the end of year 2, and \$3,500 at the end of year 3. If the discount rate is 6% , will this health care project be approved using CBA? Would your answer change if the discount rate is : a) 5% b) 4% .

3. (a) Why economic evaluation is important?
(b) What does economic evaluation mean?
(c) Give an overview of cost minimization analysis(CMA).

4. (a) Define cost effectiveness analysis(CEA).
(b) How are the effectiveness data obtained?
(c) Should effects occurring in the future be discounted?

5. (a) How does cost utility analysis(CUA) differ from CEA?
(b) When should CUA be used?
(c) Explain the methods of measuring health state utilities.

CEA

Answer any 5 (Five) of the following questions

N.B. All parts of a particular question must be answered consecutively

Using a supply and demand graph and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following: [4+3+4+3]

- A technological change that reduces the cost of producing X-rays on the market for physician services.
- Increased graduations of new doctors on the market for physician services.
- The vertical elimination of smoking in the population on the market for hospital services.
- Increased cost of medical education on the market for physician services.

- Explain the health investment function and home good function in the light of Grossman model. [8]
- How would you explain the potential use of a person's time by the labor-leisure trade-off? [3]
- Explain the impact of an increase in activities to improve health on income and leisure. [3]

Suppose Andrew's annual income is \$40,000. He can spend this amount on health care visits or on groceries (standing for all other goods) or a combination of both. The cost of a visit is \$80 and the price of a bag of grocery is \$100. [2+2+2+2+2+4]

- Make a table showing at least five consumption bundles (including two extreme consumption bundles) that Andrew can afford.
- Draw a budget line.
- Suppose Q_v is the quantity of visit and Q_g is the quantity of grocery. Write down the budget equation.
- Define opportunity cost. What is the opportunity cost of having fifteen additional visits?
- Using indifference curve, show Andrew's optimum if he buys 200 bags of groceries per year?
- Suppose that Andrew's income rises to \$48,000 per year, and that he increases his consumption of health care visits by fifty units. Using the same graphs, draw the new equilibrium. What is his income elasticity of demand for health care visits? Interpret your results.

- Define production function. Describe the production function of health. Would the production function of health eventually bend downward? Give reasons in support of your answer. [8]
- "Practitioner-provided medical interventions played a small, perhaps negligible, role in the historical decline in population mortality rates." Explain this statement with the help of empirical evidence. [6]

Suppose that the market demand for medical care is summarized by the demand function:

[14]

$$Q_d = 100 - 2p$$

and the market supply is summarized by the supply function:

$$Q_s = 20 + 2p$$

- 4-8
- Graph the demand and supply function in one diagram.
 - Calculate the equilibrium price and quantity, assuming no health insurance is available.
 - Suppose health insurance is made available that provides for a 20 percent coinsurance rate. Calculate the new equilibrium price and quantity.
 - Calculate the deadweight loss due to this insurance.

7. a) Define cost benefit analysis (CBA). Explain the basic principles of CBA.

[4]

b) What are the limitations of CBA? Explain.

[3]

10-4
c) Consider a health care project that costs \$10,000 today. It will provide benefits of \$4,000 at the end of year 1, \$3,500 at the end of year 2, and \$3,500 at the end of year 3. If the discount rate is 6%, will this health care project be approved using CBA? Would your answer change if the discount rate is: i) 5% and ii) 4%. [6]

8. a) What is economic evaluation? Why economic evaluation is important?

[4]

b) Define cost effectiveness analysis (CEA). How are the effective data obtained?

[5]

10-4
c) How does cost utility analysis (CUA) differ from CEA? When should CUA be used?

[5]